

Deniz Kılınç

Computer Engineer | Mobile & Edge AI | Embedded Systems | Full-Stack Development

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SUMMARY

Final-year computer engineering student building software end to end — from deep learning models running on-device to embedded systems on autonomous vehicles, and from Android applications to production web platforms serving real organisations. 17+ projects across five technical domains, several of them live in production. Particularly focused on edge AI, where model accuracy meets hardware constraints.

TECHNICAL SKILLS

Languages	Kotlin, Python, C++, C, TypeScript, JavaScript, PHP, Java, SQL
Mobile & Edge AI	Android, Jetpack Compose, MVVM, TensorFlow Lite, Google ML Kit, on-device inference, React Native, Expo, Flutter
AI & Computer Vision	PyTorch, TensorFlow, Keras, Vision Transformer, YOLOv8, OpenCV, MediaPipe, transfer learning, scikit-learn, CUDA
Embedded & Robotics	ROS2, NVIDIA Jetson Orin Nano, STM32 (HAL/LL), ARM Cortex-M, Arduino, I2C, SPI, UART, PWM, hardware interrupts, MQTT
Web & Backend	Next.js, React, Laravel, Node.js, Express, FastAPI, Flask-SocketIO, WebSockets, REST APIs, i18n, SSR/SSG
Data & Infrastructure	PostgreSQL, MySQL, MariaDB, Git, GitHub, Vercel, cPanel, Linux

SELECTED PROJECTS

- RhinoAI — Android application, on-device AI** 2026
Kotlin · Jetpack Compose · TensorFlow Lite · ML Kit · Vision Transformer
 - Built a facial proportion analysis app running 468-point landmark detection and deep learning inference entirely on-device at 30+ FPS.
 - Chose on-device TFLite inference over a cloud API so that patient photographs never leave the phone, eliminating both a privacy risk and server cost.
 - Converted a Vision Transformer model to TFLite with quantisation to fit mobile memory and latency budgets. Released as open source.
- Autonomous Unmanned Ground Vehicle** 2025 – 2026
ROS2 · YOLOv8 · NVIDIA Jetson Orin Nano · CUDA · OpenCV · Python · C++
 - Developed the perception and autonomous driving layer for a UGV within a university technology development society.
 - Ran CUDA-accelerated YOLOv8 on a Jetson Orin Nano instead of a lightweight CPU model, keeping detection accuracy on small obstacles at high frame rates within the vehicle's power budget.
 - Architected perception, decision and motion as separate ROS2 nodes over low-latency message channels; field-tested on demanding terrain.
- Time to Work International B.V. — corporate web platform** 2026
Next.js · TypeScript · Tailwind CSS · Framer Motion · i18n
 - Delivered a multilingual production platform for a Netherlands-based international employment company.
 - Chose SSR/SSG over a client-side SPA to maximise search indexing and first-load performance, both critical for an international audience.
 - Built the i18n layer so each page ships only its own language, avoiding the page-weight cost of loading all translations client-side.
- FocusPulse — privacy-first attention tracking (Android)** 2026
Kotlin · TensorFlow Lite · ML Kit · Python · OpenCV
 - Built an Android app measuring blink frequency, head pose and gaze direction to compute a live distraction index, processed entirely on-device.
 - Prototyped models in Python and converted them to TFLite for native performance.
 - Designed around explainable behavioural signals rather than a single black-box classification.
- Hybrid ViT-BiGRU Classification Pipeline** 2026
PyTorch · TensorFlow · CUDA · NumPy · Pandas
 - Combined Vision Transformer spatial feature extraction with bidirectional GRU sequence modelling for temporal visual data.
 - Selected BiGRU over LSTM to cut parameter count, training time and memory use while retaining bidirectional context.
 - Achieved measurable accuracy and F1-score improvements over CNN-LSTM baselines on complex temporal datasets.

TeknoHygiene — commercial hotel operations system

2026

Laravel · PHP · MariaDB · Blade · QR

- Built a QR-based, multilingual hotel housekeeping system with four permission layers (admin, manager, staff, guest), sold to hotels and configured per installation.
- Chose QR codes over NFC or handheld terminals so hotels need no hardware investment; staff scan with their own phones.
- Live in production at teknohygiene.com.

Anadolu Diabetes School — multi-module e-learning platform

2026

Laravel · PHP · MySQL · Bunny Stream · cPanel

- Built a platform combining public pages, a student panel, course sales with cart and order flow, an admin panel and a separately authenticated module.
- Solved a hard hosting constraint — shared hosting disallowed video — by routing uploads directly from the browser to Bunny Stream, so the server carries neither storage nor bandwidth load.
- Live in production at anadoludiyabetokulu.com.

Full case studies for 17 projects, including the reasoning behind each technical decision, at denizkilinc.dev

EXPERIENCE

Software Developer

2026

Confidential client — Türkiye

- Sole developer responsible for three client projects end to end: a hotel operations product, a healthcare e-learning platform and a corporate website.
- Handled database design, backend, frontend, deployment and production troubleshooting including DNS, SSL, character encoding and hosting migration.

Embedded Systems Developer

2025 – 2026

Turuncu Bilişim — Türkiye

- Wrote register- and driver-level firmware on STM32 (HAL/LL) and Arduino.
- Built communication drivers for SSD1306 OLED displays and sensor modules over I2C, SPI and UART.
- Replaced polling loops with hardware interrupts and timer-based periodic reads, freeing the CPU and capturing signals without loss.

AI4Change — Social Impact AI Project

Kodluyoruz Programme

- Applied data analytics and machine learning to civil society and social impact problems: feature engineering, predictive modelling and decision-maker-facing visualisation.

ADDITIONAL PROJECTS

Füzyon Akademi (corporate platform with custom CMS, live) · **Smart Waste Management** (IoT backend, MQTT publish/ack pipeline) · **Framecraft** (browser-based 2D rigging and animation editor) · **CodeLens AR** (code scanning through augmented reality) · **Real-Time Backend Services** (WebSockets, FastAPI, Flask-SocketIO) · **ResNet50 vs VGG19 Benchmark** (transfer learning comparison) · **Inventory Tracking App** (React Native, Expo) · **User Management System** (PHP, RBAC, secure session handling) · **2D Wizard Battle** (custom game loop and physics engine in Java)

Source code for selected projects: github.com/denizkilinc2

EDUCATION

B.Sc. Computer Engineering — Final year

Expected June 2027

Kahramanmaraş Sütçü İmam University — Türkiye

LANGUAGES

Turkish — Native · **English** — Professional working proficiency